

Part I : multilevel paradigm

- Second order/preconditioned non-smooth ML optimization
- Fixed-point multilevel schemes
- Link between BCD and ML methods
- Riemannian multilevel methods
- Multilevel methods with momentum
- Convergence rates of ML methods

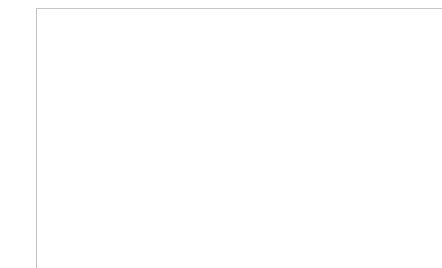
**Part II :
Image
restoration**

**Part III :
Learning
Problems**

**Part V :
Arithmetic
precision**

**Part IV :
Matrix
factorization**

- Multiresolution block PnP strategies
- Multilevel Preconditioned PnP
- Multilevel diffusion-based methods
- Unrolled multilevel methods
- Solution of ill-posed inverse problems with PINNs
- Multilevel natural gradient training of PINNs



- Mixed precision stochastic optimization
- Mixed precision domain decomposition for PINNs
- Static mixed precision accumulation
- Error analysis for transformers

- Rescaling invariant quantization of low-rank matrices
- Rescaling invariant quantization of neural networks
- Quantized butterfly factorizations for neural networks